JEDI ONBOARDING
SC24 TUTORIAL *SESSION 1B*

17 November 2024 | Andreas Herten | Jülich Supercomputing Centre, Forschungszentrum Jülich

Accessing JEDI

- Everything listed on GitHub repo of tutorial: https://go.fzj.de/mg-gh[1]

[1] Unshortened link: https://github.com/FZJ-JSC/tutorial-multi-gpu/

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1. Create JSC account at JuDoor
2. Join `training2446` project

→ https://go.fzj.de/mg-jd

3. Accept usage agreement
4. Wait 15 minutes
5. Access system via Jupyter 4.2
   JEDI, *training2446*, ***LoginNode***

→ jupyter-jsc.fz-juelich.de

6. Source course environment in a Jupyter Shell
   ```
   $ source $PROJECT_training2446/env.sh
   ```
7. Gather course material
   ```
   $ jsc-material-sync
   ```

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- We'll repeat the following steps in the first hands-on session

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- Swapcard

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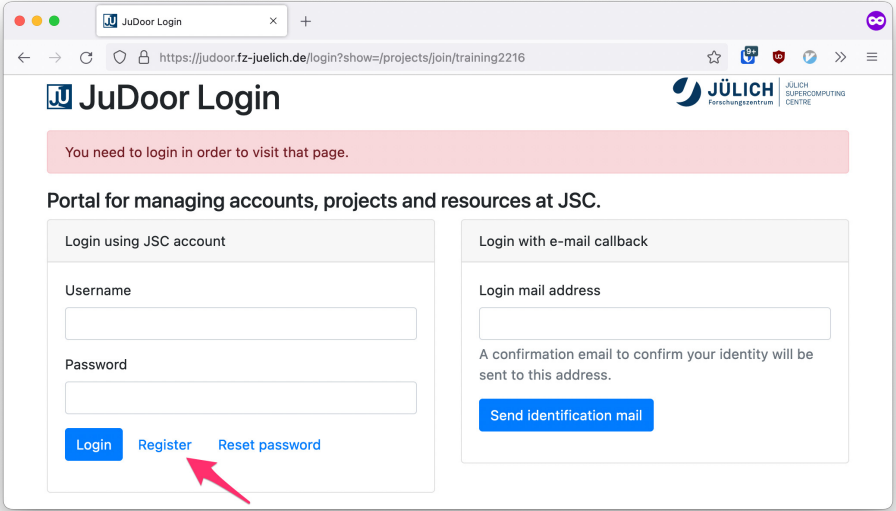
JuDoor Login
https://judoor.fz-juelich.de/login?show=/projects/join/training2216
JuDoor Login
JÜLICH Forschungszentrum
JÜLICH SUPERCOMPUTING CENTRE
You need to login in order to visit that page.
Portal for managing accounts, projects and resources at JSC.
Login using JSC account
Username
Password
Login
Register
Reset password
Login with e-mail callback
Login mail address
A confirmation email to confirm your identity will be sent to this address.
Send identification mail

Send join request to project

https://judoor.fz-juelich.de/projects/join/training2216

Your account

xyhert1

Send join request to project

JÜLICH Forschungszentrum | JÜLICH SUPERCOMPUTING CENTRE

Do you want to send a project join request to the `training2216` project?

The following information will be given to the PI and PA of the project: Dr. Andreas Herten, `xyhert1`, `an@email.address.com`

Optional additional information for the PI and PA

I'm attending the tutorial on Multi-GPU Computing and am excited to start. LET ME IN ALREADY!

Send join request to project.

training2446

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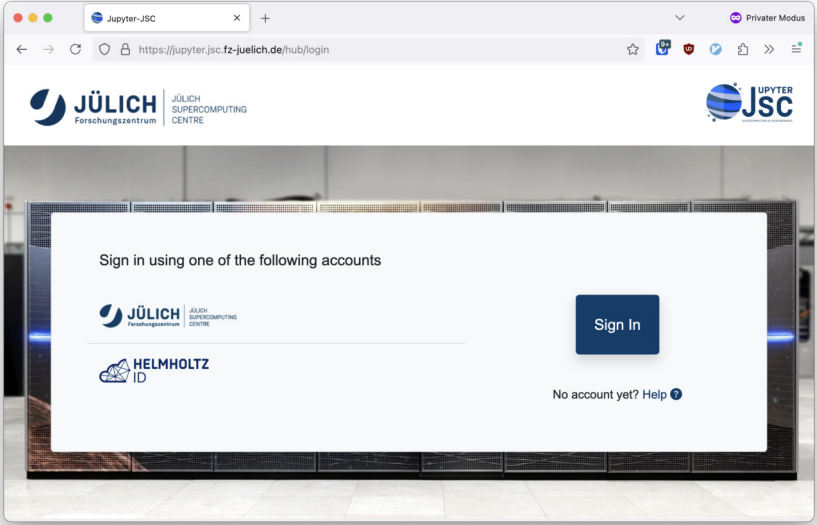
Contact Support
JuDoor Requests


Jupyter-JSC
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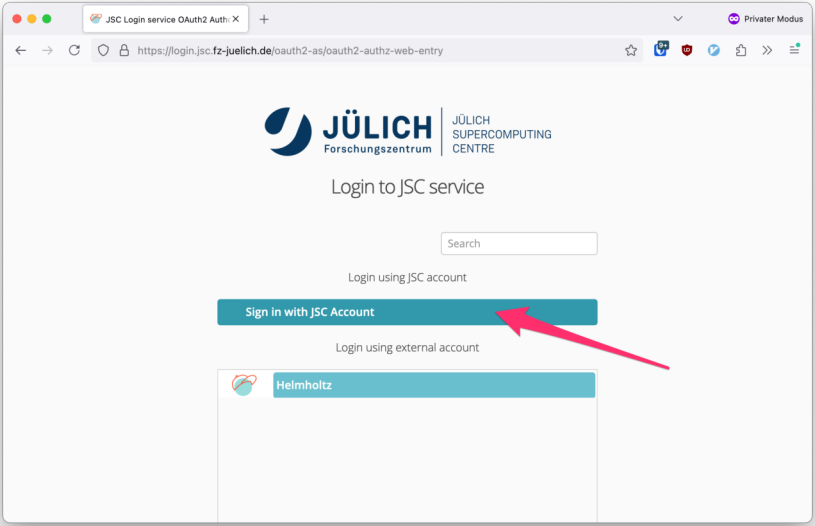
https://jupyter.jsc.fz-juelich.de/hub/login
JÜLICH
Forschungszentrum
JÜLICH
SUPERCOMPUTING
CENTRE
UPYTER
JSC
Sign in using one of the following accounts
JÜLICH
Forschungszentrum
JÜLICH
SUPERCOMPUTING
CENTRE
HELMHOLTZ
ID
Sign In
No account yet? Help


JSC Login service OAuth2 Auth
Privater Modus
https://login.jsc.fz-juelich.de/oauth2-as/oauth2-authz-web-entry
JÜLICH
Forschungszentrum
JÜLICH
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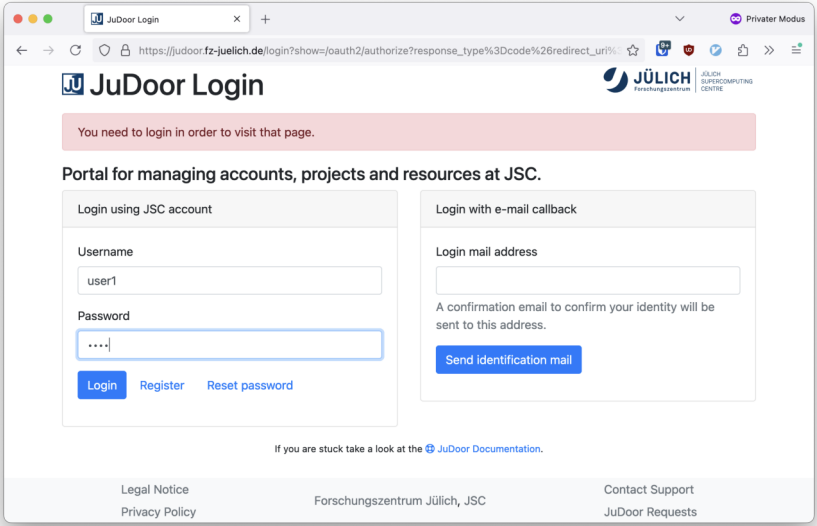
CENTRE
Login to JSC service
Search
Login using JSC account
Sign in with JSC Account
Login using external account
Helmholtz


JuDoor Login
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https://judoor.fz-juelich.de/login?show=/oauth2/authorize?response_type%3Dcode%26redirect_uri%
JuDoor Login
JÜLICH Forschungszentrum
JÜLICH SUPERCOMPUTING CENTRE
You need to login in order to visit that page.
Portal for managing accounts, projects and resources at JSC.
Login using JSC account
Username
user1
Password
Login
Register
Reset password
Login with e-mail callback
Login mail address
A confirmation email to confirm your identity will be sent to this address.
Send identification mail
If you are stuck take a look at the JuDoor Documentation.
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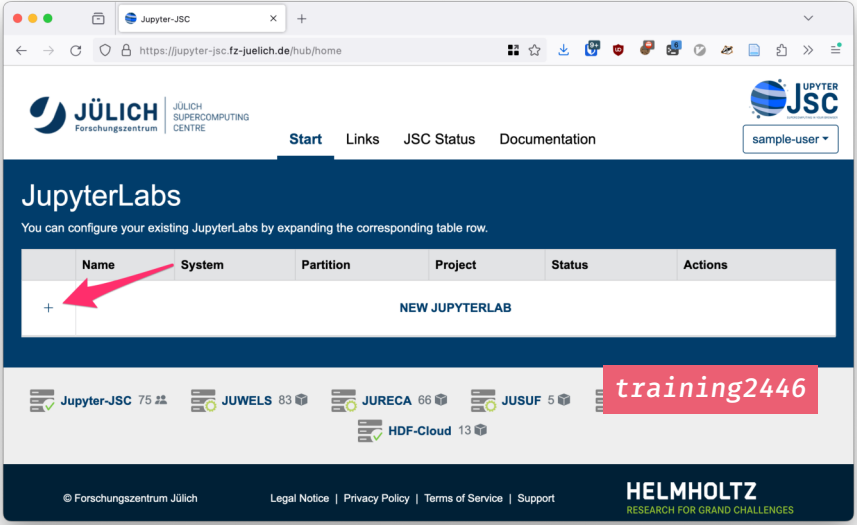
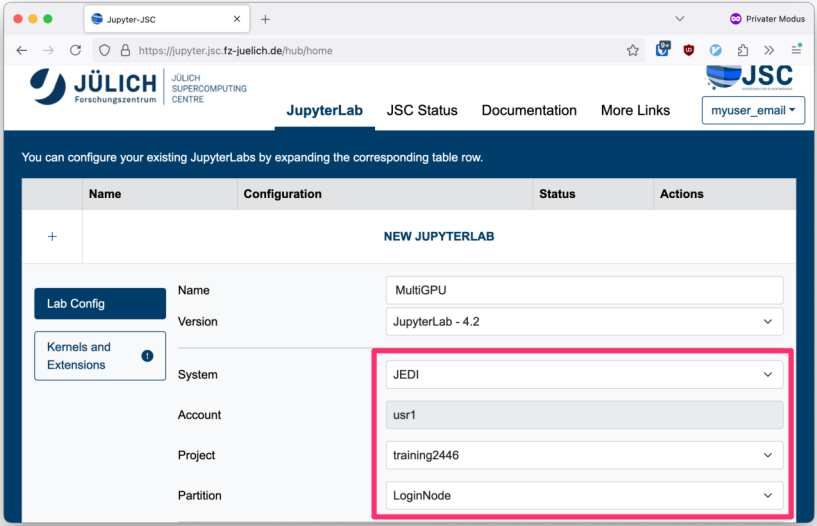
Contact Support
JuDoor Requests



go.fzj.de/mg-jd and jupyter-jsc.fz-juelich.de


Jupyter-JSC
Privater Modus
https://jupyter.jsc.fz-juelich.de/hub/home
JÜLICH Forschungszentrum
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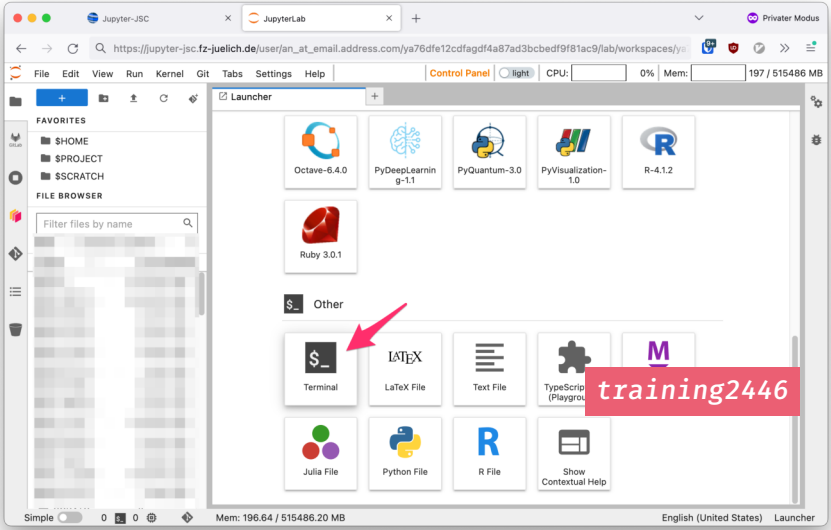
JSC
JupyterLab
JSC Status
Documentation
More Links
myuser_email
You can configure your existing JupyterLabs by expanding the corresponding table row.
Name
Configuration
Status
Actions
NEW JUPYTERLAB
Lab Config
Kernels and Extensions
Name
MultiGPU
Version
JupyterLab - 4.2
System
JEDI
Account
usr1
Project
training2446
Partition
LoginNode


Jupyter-JSC
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Privater Modus
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Control Panel
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197 / 515486 MB
FAVORITES
$HOME
$PROJECT
$SCRATCH
FILE BROWSER
Filter files by name
Launcher
Octave-6.4.0
PyDeepLearning-1.1
PyQuantum-3.0
PyVisualization-1.0
R-4.1.2
Ruby 3.0.1
Other
Terminal
LaTeX File
Text File
TypeScrip
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training2446
Julia File
Python File
R File
Show Contextual Help
Simple
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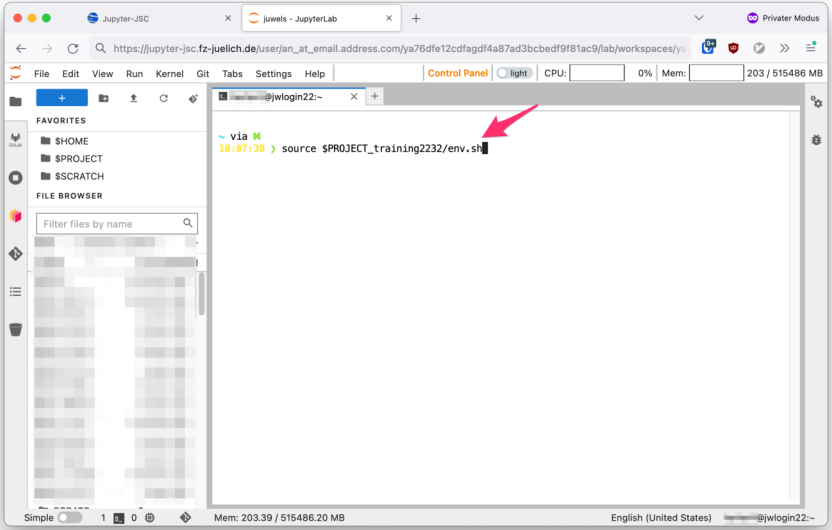
Mem: 196.64 / 515486.20 MB
English (United States)
Launcher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juwels - JupyterLab
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File Edit View Run Kernel Git Tabs Settings Help
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FAVORITES
$HOME
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10:07:30 ❯ source $PROJECT_training2232/env.sh
Simple
Mem: 203.39 / 515486.20 MB
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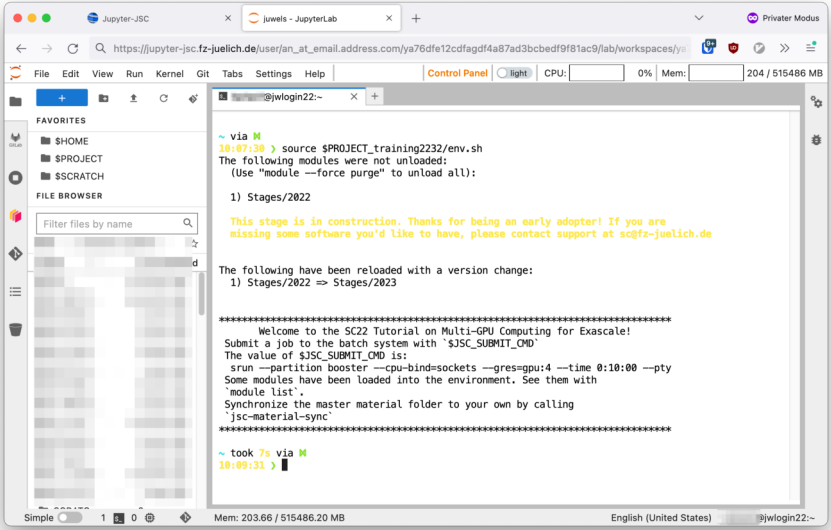
@jwlogin22:~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File Edit View Run Kernel Git Tabs Settings Help
Control Panel
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CPU:
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Mem:
204 / 515486 MB
FAVORITES
$HOME
$PROJECT
$SCRATCH
FILE BROWSER
Filter files by name
@jwlogin22:~
~ via M
10:07:30 ❯ source $PROJECT_training2232/env.sh
The following modules were not unloaded:
(Use "module --force purge" to unload all):
1) Stages/2022
This stage is in construction. Thanks for being an early adopter! If you are
missing some software you'd like to have, please contact support at sc@fz-juelich.de
The following have been reloaded with a version change:
1) Stages/2022 => Stages/2023
****
Welcome to the SC22 Tutorial on Multi-GPU Computing for Exascale!
Submit a job to the batch system with `$JSC_SUBMIT_CMD`
The value of $JSC_SUBMIT_CMD is:
srun --partition booster --cpu-bind=sockets --gres=gpu:4 --time 0:10:00 --pty
Some modules have been loaded into the environment. See them with
`module list`.
Synchronize the master material folder to your own by calling
`jsc-material-sync`
****
~ took 7s via M
10:09:31 ❯
Simple
Mem: 203.66 / 515486.20 MB
English (United States)
@jwlogin22:~

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Profiling Tools

- Extra Credits: Prepare for *Profiling Session*
- Download **Nsight Systems** now; install!

→ `https://developer.nvidia.com/nsight-systems/get-started`

- Also: Via package manager `developer.download.nvidia.com/devtools/repos`

SSH Login

SSH Login

- Login with SSH available
- We recommend Jupyter JSC: easier, more features
- Add SSH key via JuDoor to JUWELS Booster
- Important: `from` clause (limits connections to be from defined sources)
- Example

```
from="80.146.183.0/24" ssh-ed25519 AddddACadsfzaC1lZDI1NTE5AAAAsa
# coarser: from="80.144.0.0/13"
```

→ SSH: `ssh user1@juwels-booster.fz-juelich.de`

- Help at `apps.fz-juelich.de/jsc/hps/juwels/access.html`

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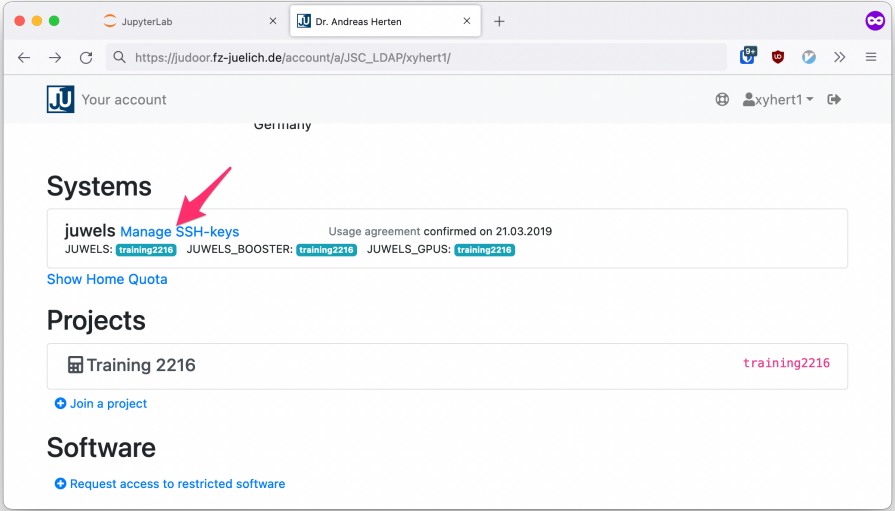
JupyterLab
Dr. Andreas Herten
https://judoor.fz-juelich.de/account/a/JSC_LDAP/xyhert1/
Your account
xyhert1
Germany
Systems
juwels Manage SSH-keys
Usage agreement confirmed on 21.03.2019
JUWELS: training2216 JUWELS_BOOSTER: training2216 JUWELS_GPUS: training2216
Show Home Quota
Projects
Training 2216
training2216
Join a project
Software
Request access to restricted software

1 - JupyterLab | SSH keys on juwels

https://judoor.fz-juelich.de/account/a/JSC_LDAP[redacted]/system/juwels/add_ssh_key

Your account

xyhert1

Upload SSH public keys

To use our systems your public key options have to include a `from=`-clause to restrict the usage of the key to your personal IP address range.
Your current IP address is `46.183.103.8`. See **the documentation** for more information.

☐ Remove all other existing public keys.

Your public key and options string

```
from="46.183.103.8" ssh-ed25519
AddddACadsfzaC1lZDl1NTE5AAAAsadf5yDS3Sht52425D0gV0AWzu52hnxilO92Ynksadfijr3bDq
```

Paste the content of your `.pub`-file here or upload a file below.

Your public key file

Browse

Additional public key options

e.g. from="46.183.103.8",...